

AuroraGrid Fund

AI-Driven Energy Platform for Data Centers

Challenges

Our project addresses the challenge of **high energy consumption and carbon emissions in data centers**, which are **growing rapidly** due to increasing demand for cloud computing and AI workloads. Currently, **data centers operate at full capacity regardless of local electricity demand or renewable energy availability**, leading to **inefficiencies, higher costs, and reliance on fossil fuels**.

A major obstacle is the **lack of coordination among independent data centers**, which prevents **dynamic workload shifting** based on real-time energy conditions. Additionally, **cloud providers may be reluctant to share data or collaborate**, and **regulatory barriers** restrict where workloads can be processed.

Our goal is to **optimize data center operations** by **leveraging AI and real-time electricity data** to shift workloads to regions with **lower electricity demand and higher renewable energy availability**, ultimately **reducing costs and carbon footprints while improving grid efficiency**.

Targets

Opportunities: **Virginia**, the largest data center market in the U.S., hosts over 35% of the world's hyperscale data centers, making it an ideal testing ground for dynamic energy optimization. Its proximity to key internet exchange points and renewable energy sources supports large-scale deployment. While tax incentives drive continued expansion, the growing energy demands of data centers are **straining the grid**, highlighting the urgent need for efficiency and load management solutions.

Size of Accessible Market: The proposal targets **smaller, independent cloud providers** in Virginia, who are more receptive to innovation and energy cost savings than hyperscale providers like AWS or Google Cloud. With multiple data centers across the state, they offer a controlled **pilot environment** and **lower barriers to collaboration**, enabling faster implementation of AI-driven energy optimization. Their need for reduced operational costs makes them ideal early adopters.

Scalability: Once successful in Virginia, the platform can scale **nationwide** by targeting **larger providers** like AWS, Google Cloud, and Microsoft Azure with proven cost and sustainability benefits. Expansion can extend to **key U.S. data center hubs**, including Dallas-Fort Worth (grid interconnections), Silicon Valley (renewable investments), Chicago (major internet exchange), and Phoenix (solar energy potential), ensuring broader impact.

How it works

Our project uses **AI-driven workload optimization** to dynamically shift computing tasks across **independent data centers** based on **real-time electricity demand and renewable energy availability**. Instead of operating at full capacity 24/7, data centers will **adjust their workloads** to regions where **energy is cheaper, grid congestion is lower, and clean energy sources like solar and wind are more abundant**.

We integrate with **real-time electricity data platforms** (e.g., Electricity Maps) to track **grid conditions, pricing, and carbon intensity**, allowing AI to **automatically route workloads to the most energy-efficient locations**. This approach **reduces energy costs, decreases carbon emissions, and enhances overall grid stability**, benefiting both cloud providers and energy markets.

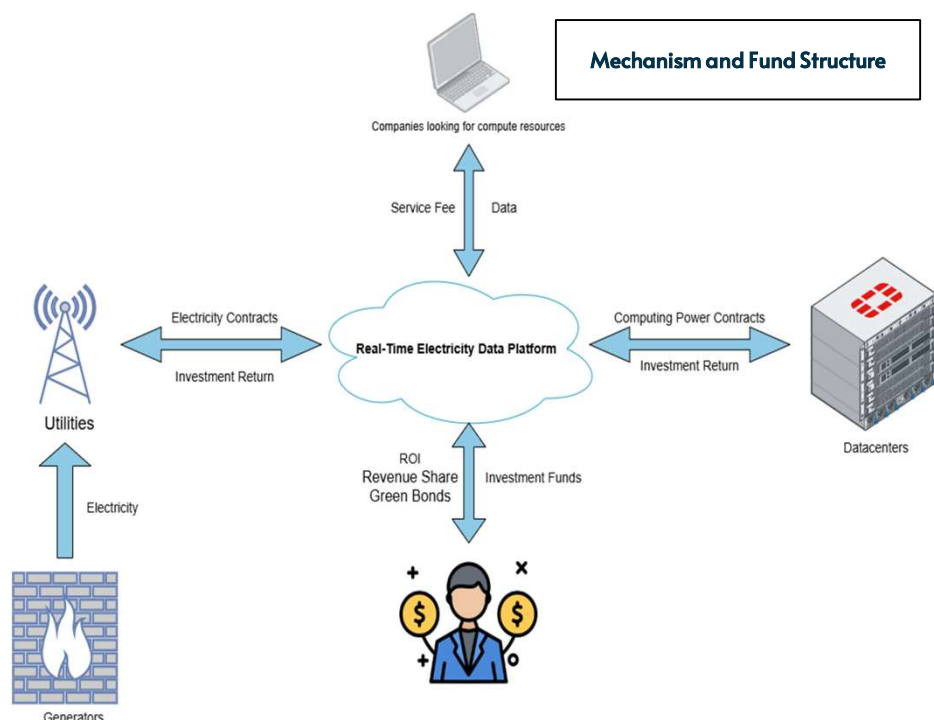
Time Horizon

Short-Term (1-2 years) – Pilot program with smaller cloud providers in Virginia.

Mid-Term (3-5 years) – Expansion to larger cloud providers and multi-state scaling.

Long-Term (5+ years) – Full commercialization, national scaling, and integration into hyperscale cloud networks.

Due Diligence Process: Our due diligence process ensures that data centers uphold environmental justice and sustainability by preventing excessive electricity demand that could strain local communities. We measure their energy consumption impact, ensuring that data centers optimize energy procurement from flexible, efficient utilities, thereby reducing grid stress and improving overall sustainability.



Operating Structure

Initial Phase: Involving a \$1M investment deployed over two years to build the core infrastructure and capabilities. These funds are mainly allocated to Platform and Software Development. Additionally, annual legal consulting for contracts and advice, targeted digital advertising, event sponsorship marketing, and transaction costs such as payment processing fees are covered.

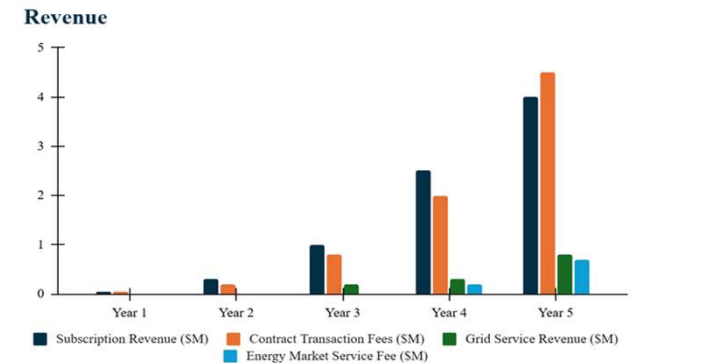
Second Phase: Raising an additional \$7M over a five-year period to support further scaling and operational growth.

Revenue Structure

Sources	Description
Subscription Based Model	Monthly Subscription Fee
Transaction Fees	Power Purchase Agreements; 1% commission
Grid Services Revenue Sharing	Automate Demand-side Response
Service Fees	Real-time Energy Market Intelligence

Capital Structure

Stage/Type	First	Second
Fund Type	Grants, Climate Tech Funds, Energy VCs	Equity, Local Utilities
Fund Size	\$1 million	\$5 million
Fund Life	7 years	
Asset Class	Convertible Bond, Grant, Venture Debt	Equity, Revenue Share
Expected IRR	12%	18%



Potential Utility and Data Center Partners

DIGITAL REALTY, EQUINIX, IRON MOUNTAIN®, Dominion Energy®, NWECC

Potential Investors

ENERGY IMPACT PARTNERS, aramco, Blackstone, BAIN & COMPANY

Risk/Challenge

Data Sharing & Competition: Large cloud providers may resist collaboration due to competitive and security concerns.

Regulatory & Compliance Issues: Some workloads (finance, healthcare, government) have strict location requirements.

Latency & Performance Concerns: Moving workloads across regions may introduce delays.

Financial Viability & Scaling: Uncertain adoption by data centers and difficulty securing initial funding.

Mitigation

Start with smaller cloud providers; ensure **data privacy controls** and allow providers to retain sensitive workloads.

Focus on **non-sensitive, compute-heavy tasks** (e.g., AI training) for workload shifting.

Prioritize **batch processing tasks**, leverage **high-speed networks (5G, edge computing)** to minimize latency.

Start with a **subscription-based model** and **revenue-sharing from energy cost savings**. Offer **early adopter incentives**, secure **ESG grants and sustainability funding**, and demonstrate **ROI through pilot projects** to attract larger cloud providers.

Net Cash Flow



Environmental & Social Impact

Environmental

Reduce carbon emissions
Reduce dependence on fossil fuel energy by shifting computing tasks to areas where renewable energy is plentiful.

Optimize grid load to reduce power waste
Reduce additional stress on the grid during peak periods through peak shaving and valley filling strategies.

Reduce water consumption
This project could reduce water consumption in data centers by reducing the reliance on highly loaded data centers.

Social

Promote the green transformation of the data center industry and create more jobs
Promote the use of more efficient energy sources in the data center industry. Encourage energy companies to develop smart grids and distributed energy management systems.

Promote ESG investment and enhance corporate social responsibility (CSR)
Help companies improve their ESG performance and access more sustainable investment funds.

